Teaching a Learning Human with One Feature per Round: Complete Solution for $K = 2$ Actions and Gaussian States

Abstract

An algorithm repeatedly shows a human decision-maker *one* feature of a randomly drawn state. The human chooses between two actions using her current (possibly wrong) beliefs about the utilities, and she learns about a feature each time it is shown. The algorithm wants to minimize the discounted sum of the human's decision regret. We solve this problem completely when the state is standard Gaussian. The key facts are: (i) the expected regret in a round depends only on the *sign* of the human's belief about the feature shown; (ii) this sign flips at most once, after a computable number τ_j of showings; (iii) hence the optimal schedule shows a *single* feature forever, namely the maximizer of an explicit score $s_j = |d_j|\,(2\delta^{\tau_j} - 1)$. All proofs are included and are self-contained.

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1 The problem

1.1 Setting

There are two actions, $k \in \{1, 2\}$, and $n \geq 1$ features. The ground-truth utilities are given by a matrix $U \in \mathbb{R}^{2\times n}$ with rows $u_1, u_2 \in \mathbb{R}^n$: in a state $x \in \mathbb{R}^n$, action k has utility $u_k^\top x$. The *regret* of choosing action y in state x is

$$\ell(x, y) = \max_{k\in\{1,2\}} u_k^\top x - u_y^\top x \ \geq 0.$$

The human holds a belief matrix $H^{(t)} \in \mathbb{R}^{2\times n}$ (rows $h_1^{(t)}, h_2^{(t)}$) at round $t = 0, 1, 2, \ldots$ A discount factor $\delta \in (0, 1)$ is fixed. A *state* is drawn fresh each round: $x^{(t)} \sim \mathcal{N}(0, I_n)$, independent across rounds and independent of everything that happened earlier.

1.2 One round

In round t:

1. The algorithm $\mathcal{A}$ chooses a feature $j_t \in [n]$ (*before* $x^{(t)}$ is drawn). Both actions are always shown.
2. $x^{(t)}$ is drawn. The human observes only the single coordinate $x_{j_t}^{(t)}$.
3. The human predicts $\hat{y}_t \in \arg\max_{k\in\{1,2\}} h_{k,j_t}^{(t)} x_{j_t}^{(t)}$, i.e. she estimates the utility of action k using only the shown feature. If the two estimates are equal, she picks one of the two actions uniformly at random, independently of everything else. She suffers regret $\ell(x^{(t)}, \hat{y}_t)$.
4. Her beliefs are updated according to the learning rule below.

1.3 Learning rule

Let

$$m_j(t) = \big|\{s < t : j_s = j\}\big|$$

be the number of times feature j has been shown *before* round t. The human's beliefs follow

$$H_{kj}^{(t)} - U_{kj} \ = \ \psi\big(m_j(t)\big)\,\big(H_{kj}^{(0)} - U_{kj}\big) \qquad \text{for all } k \in \{1, 2\},\ j \in [n],\ t \geq 0, \tag{1}$$

where $\psi : \mathbb{N} \to [0, 1]$ satisfies

$$\psi(0) = 1, \qquad \psi \text{ non-increasing}, \qquad \psi(m) \to 0 \ (m \to \infty). \tag{2}$$

Remark 1 (Relation to a squared-error rule). A "ϕ-convergent" learner is usually specified by $(U_{kj} - H_{kj}^{(t)})^2 = \phi(m_j(t))\,(U_{kj} - H_{kj}^{(0)})^2$. That only fixes the *absolute value* of the error. Rule (1) is the case $\psi = \sqrt{\phi}$ in which the error also keeps its initial sign (belief moves monotonically toward the truth, never overshooting). Writing the rule *without* absolute values is exactly the extra modeling assumption that makes the learner's state a deterministic function of the counts.

1.4 Objective

For a (possibly adaptive or randomized) rule for choosing $(j_t)_{t\ge 0}$, its cost is

$$\mathbb{E}\Big[\sum_{t=0}^{\infty} \delta^t \, \ell\big(x^{(t)}, \hat{y}_t\big)\Big]. \tag{3}$$

The problem is to minimize (3). We also consider the variant in which $\mathcal{A}$ may show *nothing* in a round ($F_t = \emptyset$); then both estimated utilities equal 0, there is a tie, and the human guesses uniformly. We call the two versions the problem with $|F_t| = 1$ and the problem with $|F_t| \le 1$.

2 Reduction to difference vectors

Define the true and perceived utility differences

$$d = u_1 - u_2 \in \mathbb{R}^n, \qquad g^{(t)} = h_1^{(t)} - h_2^{(t)} \in \mathbb{R}^n, \qquad g^{(0)} = h_1^{(0)} - h_2^{(0)}.$$

We assume $d \neq 0$ (otherwise the two actions are always equally good and the regret is identically 0).

Lemma 1 (Reduction)**.** *(a) The perceived difference satisfies $g_j^{(t)} = g_j(m_j(t))$ where*

$$g_j(m) = d_j + \psi(m)\big(g_j^{(0)} - d_j\big) = \big(1 - \psi(m)\big) d_j + \psi(m)\, g_j^{(0)}. \tag{4}$$

(b) If feature j is shown in round t, the human predicts action 1 if $g_j(m_j(t))\, x_j^{(t)} > 0$, action 2 if it is < 0, and guesses uniformly if it is $= 0$.

(c) The regret of a prediction is $\ell(x, \hat{y}) = |d^\top x| \cdot \mathbf{1}[\hat{y} \neq y^*(x)]$, where $y^*(x) = 1$ if $d^\top x > 0$ and $y^*(x) = 2$ if $d^\top x < 0$.

Proof. (a) Subtract rule (1) for $k = 2$ from the one for $k = 1$: $(H_{1j}^{(t)} - H_{2j}^{(t)}) - (U_{1j} - U_{2j}) = \psi(m_j(t))\big[(H_{1j}^{(0)} - H_{2j}^{(0)}) - (U_{1j} - U_{2j})\big]$, i.e. $g_j^{(t)} - d_j = \psi(m_j(t))(g_j^{(0)} - d_j)$.

(b) The human compares $h_{1j}x_j$ with $h_{2j}x_j$, and $h_{1j}x_j - h_{2j}x_j = g_j x_j$.

(c) Since $u_1^\top x - u_2^\top x = d^\top x$, the best action is 1 if $d^\top x > 0$ and 2 if $d^\top x < 0$. If $\hat{y} = y^*$ the regret is 0; otherwise it is $|u_1^\top x - u_2^\top x| = |d^\top x|$. (The event $d^\top x = 0$ has probability 0.) □

Thus the entire instance is described by d, $g^{(0)}$, ψ, δ.

3 The expected regret in one round

For a real number β (the human's current belief about the shown feature j), let $\Lambda_j(\beta)$ denote the expected regret in a round in which feature j is shown and the human predicts action 1 if $\beta x_j > 0$, action 2 if $\beta x_j < 0$, and guesses uniformly if $\beta = 0$; the expectation is over $x \sim \mathcal{N}(0, I_n)$ and the guess. Let $\Lambda_\emptyset$ be the expected regret when nothing is shown (a uniform guess).

3.1 A Gaussian integral

Let φ and Φ denote the standard normal density and cumulative distribution function.

Lemma 2. *Let (U, V) be jointly Gaussian with $U, V \sim \mathcal{N}(0,1)$ and correlation $r \in [-1,1]$. Then*

$$\mathbb{E}\big[\,|U|\,\mathbf{1}[UV<0]\,\big] = \frac{1-r}{\sqrt{2\pi}}.$$

Proof. Case $r = 1$. Then $V = U$ a.s., so $UV = U^2 \geq 0$ and the indicator is 0 a.s.; the value is $0 = (1-1)/\sqrt{2\pi}$.

Case $r = -1$. Then $V = -U$ a.s., so $UV < 0$ a.s. and the value is $\mathbb{E}|U| = \sqrt{2/\pi} = 2/\sqrt{2\pi} = (1-(-1))/\sqrt{2\pi}$.

Case $|r| < 1$. Write $V = rU + \sqrt{1-r^2}\,W$ with $W \sim \mathcal{N}(0,1)$ independent of U, and set $a = r/\sqrt{1-r^2}$. For $u > 0$, $V < 0$ iff $W < -au$, which has probability $\Phi(-au)$. For $u < 0$, $V > 0$ iff $W > -au$, which has probability $1 - \Phi(-au) = \Phi(au) = \Phi(-a|u|)$. Hence $\mathbb{P}(UV < 0 \mid U = u) = \Phi(-a|u|)$ and, by symmetry of the integrand,

$$\mathbb{E}\big[|U|\mathbf{1}[UV<0]\big] = \int_{-\infty}^{\infty} |u|\varphi(u)\Phi(-a|u|)\,du = 2I, \qquad I := \int_0^{\infty} u\varphi(u)\Phi(-au)\,du.$$

Integrate by parts with $dv = u\varphi(u)\,du$ (so $v = -\varphi(u)$, because $\varphi'(u) = -u\varphi(u)$) and $f(u) = \Phi(-au)$ (so $f'(u) = -a\varphi(au)$):

$$I = \Big[-\varphi(u)\Phi(-au)\Big]_0^{\infty} - \int_0^{\infty} (-\varphi(u))(-a\varphi(au))\,du = \varphi(0)\Phi(0) - a\int_0^{\infty} \varphi(u)\varphi(au)\,du.$$

Now $\varphi(0)\Phi(0) = \frac{1}{2\sqrt{2\pi}}$, and using $\int_0^{\infty} e^{-cu^2/2}du = \frac{1}{2}\sqrt{2\pi/c}$,

$$\int_0^{\infty} \varphi(u)\varphi(au)\,du = \frac{1}{2\pi}\int_0^{\infty} e^{-(1+a^2)u^2/2}du = \frac{1}{2\sqrt{2\pi}\sqrt{1+a^2}}.$$

Therefore $I = \frac{1}{2\sqrt{2\pi}}\big(1 - \frac{a}{\sqrt{1+a^2}}\big)$. Since $1 + a^2 = \frac{1}{1-r^2}$, we have $\frac{a}{\sqrt{1+a^2}} = r$, so $I = \frac{1-r}{2\sqrt{2\pi}}$ and $2I = \frac{1-r}{\sqrt{2\pi}}$. □

3.2 The one-round loss

Proposition 1 (One-round expected regret)**.** *With $x \sim \mathcal{N}(0, I_n)$ and $d \neq 0$,*

$$\Lambda_j(\beta) = \frac{\|d\| - \operatorname{sgn}(\beta)\,d_j}{\sqrt{2\pi}} \quad (\operatorname{sgn}(0) := 0), \qquad \Lambda_{\emptyset} = \frac{\|d\|}{\sqrt{2\pi}}. \tag{5}$$

Proof. Case $\beta \neq 0$. Let $s = \operatorname{sgn}(\beta)$. Since $\mathbb{P}(x_j = 0) = 0$, the human predicts action 1 iff $s\,x_j > 0$, and by Lemma 1(c) the regret is $|d^\top x|\mathbf{1}[\text{wrong}]$. Put $U = d^\top x/\|d\|$ and $V = s\,x_j$. Both are standard normal and jointly Gaussian (linear functions of x), and since $\mathbb{E}[x_i x_j] = \mathbf{1}[i=j]$,

$$r = \operatorname{Corr}(U,V) = \mathbb{E}[UV] = \frac{s\,d_j}{\|d\|}.$$

The human is wrong iff $\operatorname{sgn}(V) \neq \operatorname{sgn}(U)$, i.e. $UV < 0$ (a.s.), and the regret equals $\|d\|\,|U|\mathbf{1}[UV<0]$. By Lemma 2,

$$\Lambda_j(\beta) = \|d\|\,\frac{1-r}{\sqrt{2\pi}} = \frac{\|d\| - s\,d_j}{\sqrt{2\pi}}.$$

Case $\beta = 0$ and the case of showing nothing. The human guesses uniformly and independently of x, so she is wrong with probability $\frac{1}{2}$ regardless of x, and the expected regret is $\frac{1}{2}\mathbb{E}|d^\top x|$. Since $d^\top x \sim \mathcal{N}(0, \|d\|^2)$, $\mathbb{E}|d^\top x| = \|d\|\sqrt{2/\pi}$, and $\frac{1}{2}\|d\|\sqrt{2/\pi} = \|d\|/\sqrt{2\pi}$. This equals $\Lambda_j(0)$ given by (5) with $\operatorname{sgn}(0) = 0$, and equals $\Lambda_{\emptyset}$. □

Consequences. Fix a feature j and put

$$a_j = \frac{\|d\| - |d_j|}{\sqrt{2\pi}}, \qquad b_j = \frac{\|d\| + |d_j|}{\sqrt{2\pi}}, \qquad \Delta_j = b_j - a_j = \sqrt{\tfrac{2}{\pi}}\,|d_j|.$$

Then $\Lambda_j(\beta) = a_j$ if $\operatorname{sgn}(\beta) = \operatorname{sgn}(d_j)$ ("good" belief), $\Lambda_j(\beta) = b_j$ if $\operatorname{sgn}(\beta) = -\operatorname{sgn}(d_j)$ ("bad" belief), and $\Lambda_j(0) = \frac{a_j+b_j}{2} = \Lambda_\emptyset$. In words:

- *Only the sign of the human's belief matters, not its magnitude.* A human who overestimates a feature by a factor of 100 does exactly as well as one who is nearly right, provided the sign is right.
- A wrong-sign feature is *worse than showing nothing* ($b_j > \Lambda_\emptyset$), and a right-sign feature is *better* ($a_j < \Lambda_\emptyset$).
- The value of getting feature j's sign right is $\Delta_j = \sqrt{2/\pi}\,|d_j|$, proportional to its true importance.

4 How the human's belief evolves

We assume the following *non-degeneracy* conditions (relaxed in Remark 4): for all j,

$$d_j \neq 0, \qquad g_j^{(0)} \neq 0, \qquad \text{and, if } \operatorname{sgn} g_j^{(0)} \neq \operatorname{sgn} d_j,\ \psi(m) \neq \theta_j \text{ for all } m,$$

where θ_j is defined below.

Lemma 3 (Sign of the belief)**.** *Fix j and write $g_j(m) = (1-\psi(m))d_j + \psi(m)g_j^{(0)}$ as in (4); this is a convex combination of d_j and $g_j^{(0)}$ because $\psi(m) \in [0,1]$.*

(a) If $\operatorname{sgn} g_j^{(0)} = \operatorname{sgn} d_j$, then $\operatorname{sgn} g_j(m) = \operatorname{sgn} d_j$ for every m.

(b) If $\operatorname{sgn} g_j^{(0)} = -\operatorname{sgn} d_j$, put $\theta_j := \dfrac{|d_j|}{|d_j| + |g_j^{(0)}|} \in (0,1)$. Then

$$\operatorname{sgn} g_j(m) = \begin{cases} \operatorname{sgn} d_j, & \psi(m) < \theta_j, \\ 0, & \psi(m) = \theta_j, \\ -\operatorname{sgn} d_j, & \psi(m) > \theta_j. \end{cases}$$

Proof. (a) Both terms of the convex combination have the sign of d_j (weakly), and at least one has weight > 0 and is nonzero, so the sum has the sign of d_j.

(b) By symmetry take $d_j > 0 > g_j^{(0)}$. Then $g_j(m) = (1-\psi)\,|d_j| - \psi\,|g_j^{(0)}| > 0 \iff |d_j| > \psi\,(|d_j| + |g_j^{(0)}|) \iff \psi < \theta_j$, and similarly for $= 0$ and < 0. □

Definition 1 (Flip count)**.** Set $\tau_j = 0$ in case (a). In case (b), set $\tau_j = \min\{m \geq 0 : \psi(m) < \theta_j\}$. Because $\psi(0) = 1 > \theta_j$ and $\psi(m) \to 0$ with $\theta_j > 0$, we have $1 \leq \tau_j < \infty$.

Since ψ is non-increasing, $\psi(m) > \theta_j$ for $m < \tau_j$ (no ties) and $\psi(m) < \theta_j$ for $m \geq \tau_j$. Combining Lemma 3 with Proposition 1 gives the cost of showing feature j for the $(m+1)$-st time:

$$c_j(m) := \Lambda_j\big(g_j(m)\big) = \begin{cases} b_j, & m < \tau_j, \\ a_j, & m \geq \tau_j. \end{cases} \tag{6}$$

This is a non-increasing sequence in m (a stretch of "bad" rounds of length τ_j, then "good" forever). For the empty option set $c_\emptyset(m) \equiv \Lambda_\emptyset$, also non-increasing.

Remark 2 (Explicit τ_j for geometric learning). If $\psi(m) = \rho^m$ with $\rho \in (0,1)$, then $\psi(m) < \theta_j \iff m > \ln(1/\theta_j)/\ln(1/\rho)$, so $\tau_j = \big\lfloor \ln(1/\theta_j)/\ln(1/\rho) \big\rfloor + 1$ in case (b).

5 Reducing the objective to counts

Lemma 4 (Open-loop sequences suffice)**.** *For any deterministic sequence* $\mathbf{j} = (j_t)_{t\geq 0}$ *the cost* (3) *equals*

$$C(\mathbf{j}) = \sum_{t=0}^{\infty} \delta^t \, c_{j_t}\big(m_{j_t}(t)\big). \tag{7}$$

Moreover, for any adaptive or randomized rule the cost is at least $\inf_{\mathbf{j}} C(\mathbf{j})$ *(infimum over deterministic sequences). Hence it suffices to optimize over deterministic sequences.*

Proof. For a deterministic sequence, the counts $m_j(t)$ and hence $g_j(m_j(t))$ are deterministic by Lemma 1(a). The state $x^{(t)}$ is independent of the choice j_t and of the human's belief at time t (both are determined before $x^{(t)}$ is drawn), so the expected regret in round t is $\Lambda_{j_t}(g_{j_t}(m_{j_t}(t))) = c_{j_t}(m_{j_t}(t))$. Summing with weights δ^t (all terms are nonnegative, so exchanging expectation and sum is justified) gives (7); the sum is finite since each term is at most $b_{\max}\delta^t$.

For an adaptive or randomized rule, the sequence (j_t) is random, but the human's belief at round t depends only on the counts, hence only on $j_0, \ldots, j_{t-1}$, and $x^{(t)}$ is independent of $j_0, \ldots, j_t$. Conditioning on $(j_s)_{s\leq t}$, the expected regret in round t is again $c_{j_t}(m_{j_t}(t))$. So the cost is $\mathbb{E}[C(\mathbf{j})] \geq \inf_{\mathbf{j}} C(\mathbf{j})$. □

So the problem is now purely combinatorial: choose an infinite sequence of features, where showing feature j costs $c_j(m)$ if it has been shown m times before, and later costs are discounted.

6 The main theorem: a single feature forever

The structural result below only uses that every cost sequence is non-increasing, and so is stated in that generality.

Theorem 1 (Optimal schedule is constant)**.** *Let* $\mathcal{I}$ *be a finite index set and let* $c_i : \mathbb{N} \to [0, B]$ *be non-increasing for every* $i \in \mathcal{I}$*. For a sequence* $\mathbf{i} = (i_t)_{t\geq 0}$ *in* $\mathcal{I}$ *let* $m_i(t) = |\{s < t : i_s = i\}|$ *and*

$$C(\mathbf{i}) = \sum_{t\geq 0} \delta^t \, c_{i_t}\big(m_{i_t}(t)\big), \qquad W_i(m) := \sum_{s\geq 0} \delta^s \, c_i(m+s), \qquad J_i := W_i(0).$$

(J_i is the cost of choosing i in every round.) Then

$$\inf_{\mathbf{i}} C(\mathbf{i}) = \min_{i\in\mathcal{I}} J_i,$$

and the infimum is attained by the constant sequence $i_t \equiv i^*$ *for any* $i^* \in \arg\min_i J_i$*.*

Proof. Two facts about each W_i follow from monotonicity of c_i and the definition:

(F1) $W_i(m) = c_i(m) + \delta W_i(m+1)$.

(F2) $W_i(m+1) \le W_i(m)$ (compare the sums term by term: $c_i(m+1+s) \le c_i(m+s)$), and $c_i(m) \ge (1-\delta)W_i(m+1)$ (because $c_i(m) \ge c_i(m+1+s)$ for every $s \ge 0$ and $(1-\delta)\sum_{s\ge 0}\delta^s = 1$).

For a vector of counts $\mathbf{m} = (m_i)_{i\in\mathcal{I}}$ define

$$M(\mathbf{m}) = \min_{i\in\mathcal{I}} W_i(m_i), \qquad 0 \le M \le \frac{B}{1-\delta}.$$

Claim. For every $\mathbf{m}$ and every $j \in \mathcal{I}$,

$$M(\mathbf{m}) \le c_j(m_j) + \delta\, M(\mathbf{m} + e_j), \tag{8}$$

with equality if $W_j(m_j) = M(\mathbf{m})$.

Proof of claim. Let $R = \min_{i\ne j} W_i(m_i)$ ($R = +\infty$ if $|\mathcal{I}| = 1$). Then $M(\mathbf{m}+e_j) = \min\{W_j(m_j+1), R\}$.

Case 1: $W_j(m_j+1) \le R$. Then the right side of (8) is $c_j(m_j)+\delta W_j(m_j+1) = W_j(m_j) \ge M(\mathbf{m})$ by (F1). If moreover $W_j(m_j) = M(\mathbf{m})$ this is an equality.

Case 2: $R < W_j(m_j+1)$. The right side is $c_j(m_j) + \delta R \ge (1-\delta)W_j(m_j+1) + \delta R \ge (1-\delta)R + \delta R = R \ge M(\mathbf{m})$, using (F2) and then $W_j(m_j+1) > R$. In this case $W_j(m_j)$ is not the minimum: if it were, then $W_j(m_j+1) \le W_j(m_j) = M(\mathbf{m}) \le R$ by (F2), contradicting Case 2. So the equality statement is not needed here. This proves the claim.

Lower bound. Let $\mathbf{i}$ be any sequence and write $\mathbf{m}(t)$ for its count vectors, so $\mathbf{m}(t+1) = \mathbf{m}(t)+e_{i_t}$. Applying (8) at each step,

$$M(\mathbf{m}(0)) \le c_{i_0}(m_{i_0}(0)) + \delta M(\mathbf{m}(1)) \le \cdots \le \sum_{t<T} \delta^t c_{i_t}(m_{i_t}(t)) + \delta^T M(\mathbf{m}(T)).$$

Since $0 \le M \le B/(1-\delta)$, the last term tends to 0 as $T \to \infty$, hence $M(\mathbf{0}) \le C(\mathbf{i})$. Because $M(\mathbf{0}) = \min_i W_i(0) = \min_i J_i$, every sequence costs at least $\min_i J_i$.

Attainment. The constant sequence at $i^* \in \arg\min_i J_i$ costs exactly $J_{i^*} = \min_i J_i$. □

Remark 3. For readers who know it: this is a deterministic special case of the Gittins index theorem for multi-armed bandits (each feature is an "arm" whose state advances only when it is shown). The proof above is self-contained and does not need that theorem.

7 The solution

Apply Theorem 1 with the options $\mathcal{I} = [n]$ (and also $\emptyset$ in the variant $|F_t| \le 1$) and the non-increasing sequences (6).

Lemma 5 (Cost of a constant schedule)**.** *For every $j \in [n]$, with τ_j as in the Flip-count definition,*

$$J_j = \frac{b_j(1-\delta^{\tau_j}) + a_j\,\delta^{\tau_j}}{1-\delta} = \frac{1}{\sqrt{2\pi}(1-\delta)}\Big(\|d\| - s_j\Big), \qquad s_j := |d_j|\big(2\delta^{\tau_j} - 1\big).$$

Also $J_\emptyset = \dfrac{\|d\|}{\sqrt{2\pi}(1-\delta)}$, *which corresponds to* $s_\emptyset = 0$.

Proof. By (6), $J_j = b_j \sum_{t<\tau_j} \delta^t + a_j \sum_{t\geq\tau_j} \delta^t = b_j \frac{1-\delta^{\tau_j}}{1-\delta} + a_j \frac{\delta^{\tau_j}}{1-\delta}$. Substituting a_j, b_j:

$$b_j(1-\delta^\tau) + a_j\delta^\tau = a_j + (b_j - a_j)(1-\delta^\tau) = \frac{\|d\| - |d_j| + 2|d_j|(1-\delta^\tau)}{\sqrt{2\pi}} = \frac{\|d\| - |d_j|(2\delta^\tau - 1)}{\sqrt{2\pi}}.$$

The statement for $J_\emptyset$ is immediate from $c_\emptyset \equiv \|d\|/\sqrt{2\pi}$. □

Corollary 1 (Optimal schedule and optimal loss)**.** *Under the standing assumptions (Gaussian state, $K = 2$, learning rule (1)–(2), non-degeneracy), the optimal strategy is to show one feature in every round,*

$$\boxed{F_t = \{j^*\} \;\; \textit{for all } t \geq 0, \qquad j^* \in \arg\max_j \; s_j, \qquad s_j = |d_j|\big(2\delta^{\tau_j} - 1\big),}$$

and the optimal discounted regret is

$$\boxed{V^* = \frac{\|d\| - \max_j s_j}{\sqrt{2\pi}\,(1-\delta)}}$$

(for the variant $|F_t| \leq 1$, replace $\max_j s_j$ by $\max\{0, \max_j s_j\}$, i.e. show nothing if every score is negative). No adaptive or randomized strategy does better.

Proof. By Lemma 4 it suffices to consider deterministic sequences. By Theorem 1 the minimal cost is $\min_i J_i$, attained by a constant sequence. By Lemma 5, J_j is a decreasing function of s_j, so $\min_j J_j$ corresponds to $\max_j s_j$. □

8 Interpretation and sanity checks

Teaching vs. exploiting. A feature that is already believed with the right sign ($\tau_j = 0$) has score $s_j = |d_j|$. A feature believed with the wrong sign has to be shown τ_j times at the higher cost b_j before its belief flips; its score is $|d_j|(2\delta^{\tau_j} - 1) < |d_j|$. Consequently:

- **Half-life rule.** A wrong-sign feature is better than showing nothing iff $\delta^{\tau_j} > \frac{1}{2}$, i.e. iff the flip happens within the discount half-life $\tau_j < \ln 2 / \ln(1/\delta)$.
- **No interleaving.** Teaching is never done "a little at a time" or postponed: the optimal schedule picks its feature at $t = 0$ and never changes. (For instance, exploiting an already-correct feature i for s rounds and then switching to feature j costs $J_i(1-\delta^s) + \delta^s J_j$, which is linear in δ^s, so the best switching time is $s = 0$ or $s = \infty$.)
- **Myopic limit** $\delta \to 0$: every wrong-sign feature has a negative score, so the choice is the largest $|d_j|$ among currently right-sign features.
- **Patient limit** $\delta \to 1$: every wrong-sign feature has $s_j \to |d_j|$, so the choice is the largest $|d_j|$ among *all* features; the asymptotic per-round loss approaches $(\|d\| - \max_j |d_j|)/\sqrt{2\pi}$.

Worked example. Let $n = 2$, $d = (1, 3)$, $g^{(0)} = (2, -3)$, $\psi(m) = 0.6^m$. Feature 1 starts with the right sign, so $\tau_1 = 0$ and $s_1 = 1$. For feature 2, $\theta_2 = \frac{3}{3+3} = \frac{1}{2}$ and $\psi(1) = 0.6 > \frac{1}{2} > \psi(2) = 0.36$, so $\tau_2 = 2$ and $s_2 = 3(2\delta^2 - 1)$. Hence:

δ	s_1	s_2	show forever	V^*
0.9	1	1.86	feature 2 (teach)	5.195
0.7	1	−0.06	feature 1 (exploit)	2.875

Teaching feature 2 beats exploiting feature 1 iff $3(2\delta^2 - 1) > 1 \iff \delta > \sqrt{2/3} \approx 0.816$. Feature 2 beats showing nothing iff $\delta > 1/\sqrt{2} \approx 0.707$.

Numerical check. For this example and for random instances ($n = 3$), a brute-force value iteration over count vectors (counts capped at τ_j) gave $V^*(\mathbf{0})$ agreeing with Corollary 1 to within 10^{-6}, and Monte Carlo simulation of the one-round regret agreed with Proposition 1 to within sampling error. This is a sanity check, not part of the proofs.

9 Remarks and extensions

Remark 4 (Ties and degenerate features). The non-degeneracy assumptions are not essential.

- *Ties* ($\psi(m) = \theta_j$ for some m, or $g_j^{(0)} = 0$): at those counts the cost is the middle value $\frac{a_j+b_j}{2}$ by Proposition 1. The sequence $c_j(m)$ is still non-increasing ($b_j \geq \frac{a_j+b_j}{2} \geq a_j$ in that order, and monotone ψ preserves the order), so Theorem 1 applies unchanged; only the closed form of J_j changes (compute $J_j = \sum_t \delta^t c_j(t)$ directly).
- *Irrelevant features* ($d_j = 0$): then $g_j(m) = \psi(m) g_j^{(0)}$ has constant sign and $\Lambda_j \equiv \|d\|/\sqrt{2\pi}$, the same as showing nothing, so $s_j = 0$.
- *Never-learned features* would have $\tau_j = \infty$ and score $-|d_j|$; they are excluded by $\psi(m) \to 0$.

Remark 5 (Why $|F_t| = 1$ is easy, and what changes for $|F_t| \geq 2$). Theorem 1 used that (i) the loss in a round depends only on the count of the *one* feature shown, and (ii) counts advance only when that feature is shown. With $|F_t| \geq 2$ the human sums $g_j x_j$ over the shown features, the correlation between her perceived and true margin is $r = \frac{g_F^\top d_F}{\|g_F\|\|d\|}$ (for $x \sim \mathcal{N}(0, I_n)$), and the loss is no longer a function of any single count: magnitudes of the g_j matter, features interact, and the problem is no longer a collection of independent cost sequences. This is where the harder part of the problem begins.

Remark 6 (Other state distributions). Only Proposition 1 used the Gaussian assumption. For another distribution one only needs to recompute the one-round loss as a function of $(F, \operatorname{sgn} g_j)$; Lemma 4 and Theorem 1 apply as long as each feature's cost sequence is non-increasing in its count.